

CSE:4227 Digital Image Processing (Sec- A, B, C)

Fall 2021 (24 May – 08 Sept., 2022)

Class wise Topics/Contents: (Mon/Wed/Thurs:: each class 50 mnt)

Instructors: Kazi A Kalpoma and [Sanzana Karim Lora](#)

<u>Week/Class</u>		<u>Date</u>	<u>Topics covered</u>
<u>1</u>			
	C_1	25/05	Introduction to DIP , Course policy, outline, What is DIP, Why do we need DIP? Key stages in DIP and how they works.
	C_2	26/05	Image Sensing and Acquisition: Visual Perception and Image formation model, Image acquisition, Sensors, Image representation, Image Digitization
<u>2</u>	C_3	30/05	Spatial and Intensity Resolution: Spatial and Gray Level Resolutions, Effect of Spatial and Gray Level Resolutions,
	C_4	01/06	Pixels & their relationships: neighborhood, adjacency, connectivity, paths and regions. Distance matrix (city block, chessboard, and Euclidean distance),
	C_5	02/06	Revision and solving exercises on connectivity, paths and distance measure.
<u>3</u>	C_6	06/06	Intensity Transformations: Some basic Gray Level/Intensity Transformations: Point processing-LUT, Linear-Negative, Identical, Thresholding, Log, and Power Law.
	C_7	08/06	Image Enhancement: Piecewise Linear Transformation (Contrast Stretching, Intensity Level Slicing, Bit Plane Slicing). Exercise problems on them.
	C_8	09/06	Image Interpolation: Image Scaling (Zooming and Shrinking) process. Nearest neighbor interpolation.
<u>4</u>	C_9	13/06	Image histogram Processing: Image histogram, Use of histogram, Normalized histogram, Histogram processing techniques: histogram Sliding, Stretching histogram.
	C_10	15/06	Image histogram equalization: Need of image histogram equalization, cumulative density function (CDF) and cumulative histogram, histogram equalization algorithm, equalization transform functions

	C_11	16/06	Chap 04 (Introduction to Frequency domain)
5	C_12	20/06	Revision Class (@google Classroom)
	C_13	22/06	Quiz-1
	C_14	23/06	Chap 04 (Filtering in Frequency domain)
6	C_15	27/06	Chap 04 (Filtering in Frequency domain)
	C_16	29/06	Filtering in Spatial Domain: What is masking? Smoothing spatial filters: (mean, weighted mean filter), Noise removing, correlation & convolution. Gaussian filter,
	C_17	30/06	Non-linear Median filter, different filtering designs and their applications. Sharpening spatial Filters: Image derivatives, Laplacian Filter, Laplacian Image Enhancement, Difference filters, Combining filtering techniques.
7	C_18	04/07	Morphological Image Processing Part A: What is morphology? Basic Set Theory, Structuring Element, Basic operations: Erosion, Dilation, Applications of basic morphological operations,
	C_19	06/07	Opening, Closing Morphological Image Processing, Compound operations. Morphological Algorithms: Boundary extraction. Region filling
	C_20	07/07	Quiz-2
8		08/08	EID + MID BREAK
		23/07	
9	C_21	25/07	Morphological Image Processing Part B: Shape or Boundary detection by Hit and Miss Transform. Morphological Algorithms.
	C_22	27/07	Thinning and Thickening Morphological process
	C_23	28/07	Chap 04 (Filtering in Frequency domain)
10	C_24	01/08	Image Segmentation: Definition, segmentation problems, different techniques of segmentation, Introduction of edges, Finding points and line detection.
	C_25	03/08	Edge detection approach for segmentation. Canny edge detection and edge linking
	C_26	04/08	Line detection using Hough Transform.

<u>11</u>	C_27	08/08	Region Based Segmentation. Clustering methods, Boundary detection
	C_28	10/08	Quiz-3
	C_29	11/08	Colour Image Processing: Colour fundamentals and Colour models: RGB Color Model.
<u>12</u>	C_30	15/08	GOV. Holiday
	C_31	17/08	Image Compression: Why do we need compression? Compression Ratio and Relative Data Redundancy. How can we implement compression: Coding redundancy, Spatial and temporal redundancy, Irrelevant information.
	C_32	18/08	GOV. Holiday
<u>13</u>	C_33	22/08	Information theory review, Image Entropy, Lossy and Loss-free Compression, Different Compression Techniques: Huffman coding
	C_34	24/08	Different Compression Techniques: Golomb coding, Arithmetic coding
	C_35	25/08	Colour models: RGB Color and HSI Color Model.
<u>14</u>	C_36	29/08	Different Compression Techniques: LZW coding, Run length encoding (RLE), RLE on Binary Image, RLE on BMP.
	C_37	31/08	Quiz-4
	C_38	01/09	What is image restoration? Noise models, Noise removal using spatial domain filtering. Revision on Spatial Filters.
<u>15</u>	C_39	05/09	Image Representation: Chain Code and Image Fidelity Criteria: Subjective and Objective
	C_40	07/09	Revision
	C_41	08/09	Farewell